HumeBot — Executive Brief for University IT

HUMEBOT IN ONE SENTENCE

HumeBot is a faculty-led, evidence-grounded instructional AI resource that helps students work directly with David Hume's *Enquiry concerning Human Understanding*.

THE EDUCATIONAL PROBLEM

A general-purpose chatbot can give a fluent, confident answer about a philosophical text while misquoting it or citing passages that do not say what it claims. In a course built on close reading of a primary text, students may not be able to distinguish a reliable response from a plausible-sounding one, and instructors may have little visibility into what the answer rested on. Our own testing confirmed the problem — even carefully prompted models misquoted the text often enough that their quotations could not be treated as authoritative.

WHAT IS DIFFERENT ABOUT HUMEBOT

HumeBot answers only from a controlled copy of the text (*Enquiry* Sections I–IX). A separate step selects the passages an answer needs, and the answer is written from those passages alone.

Quotations are created by the software, not the AI model: every quotation a student sees is Hume's exact text at a location that can be checked. Automatic rules check the structure of the evidence and its provenance before anything is shown, and when the system cannot support an answer it says so visibly instead of guessing.

It is a supplement to the professor and the text — a further place to ask questions between classes — not an autonomous instructor, a grading tool or a general chatbot. Its scope is deliberately narrow so that its behaviour can be tested thoroughly, and the system is designed so faculty can inspect its responses, evidence, and failures.

EVIDENCE SO FAR

- A **60-question held-out study** on newly written questions.
- In that study, **238 of 238** of the constrained system's citations were exact, valid and supported by the evidence it selected — every quotation it showed was Hume's actual wording, at a location that exists and that the answer actually relied on.
- **1,205 automated tests**.
- A **frozen system configuration**, with an automated check that detects any change.
- A formal **200-question qualification study** already designed, with its pass criteria fixed in advance — **not yet run**.

The system produced a verified answer on **50 of 60** held-out questions. The remaining cases ended in visible non-answer states rather than being silently converted into unsupported answers. This is also why further qualification is necessary: provenance and answer availability are separate requirements.

WHY THE UNIVERSITY MATTERS NOW

We want to run that formal qualification on the deployment that could plausibly become the classroom deployment: an open-weight model whose exact version and configuration are pinned and documented, running under University access, privacy and security controls. If the system passes, the University would then have evidence about the exact system students would use — not about a different service that was later swapped in. The external research service used for development does not expose the model details needed to do that, so a University-supported environment is the next enabling step.

WHAT WE ARE ASKING FOR

A technical feasibility conversation: could University-supported infrastructure provide a small, pinned, reproducible deployment of an open-weight model, with sufficient context capacity, course-scoped access and appropriate storage for pseudonymous logs — and what constraints should shape it?

WHAT WE ARE NOT ASKING FOR

Not immediate production approval, not University-wide deployment, not a vendor commitment, not IT certification of the philosophical content, and not any processing of real student data before the University's governance review.

Next step. We would welcome a short meeting with the appropriate cloud or research-computing staff to discuss feasibility and constraints. A fuller technical and evaluation case is available on request.